

Energy Harvesting 2026

Exercise 6: Power Management

Task 1: Solar cell

You have a solar cell measuring 3.5 cm x 4.5 cm and want to know the amount of energy available on a sunny day. You have the Illumination data in [klx; kilo lux] from figure 1 and a 6th degree polynomial at your disposal. How much energy hits the solar cell on the third day of the graph? How large would a 5.5 V capacitor have to be to store this energy?

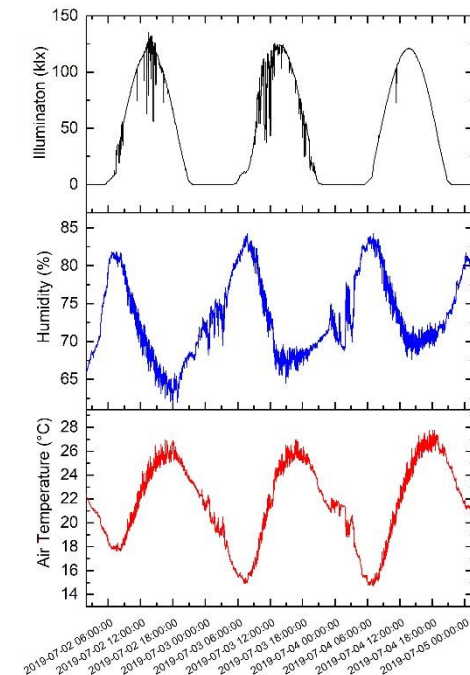


Figure 1 - Solar radiation, temperature and Air humidity over three summer days

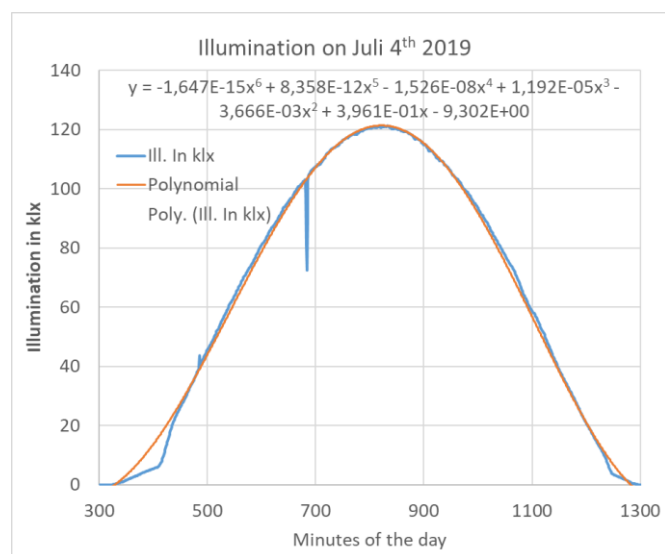


Figure 2 - Fit - 6th degree polynomial over the third day

Umrechnungsfaktoren für Messwerte üblicher Lichtmessgeräte für Sonnenlicht nach McCree (1981):

Aktueller Wert (Meßwert zum Zeitpunkt x)					
Einheit: umrechnen aus => in ...	Sensor bewertet ca.	... W/m ²	... klx	... μmol/(s·m ²) PAR	... W/m ² PAR
W/m ² Solarimeter	300-3000 nm	1	0,13	2,3	0,5
klx	380-780 nm	8	1	18	4
μmol/(s·m ²) PAR	400-700 nm	0,43	0,056	1	0,22
W/m ² PAR	400-700 nm	2	0,26	4,6	1

Figure 3 - Energy and light conversion table

Solution

Solar cell: $A = 3,5 \cdot 4,5 \text{ cm}^2 = 15,75 \text{ cm}^2 = 0,001575 \text{ m}^2$

Integral of the polynomial between 300 and 1300 min (limited to positive values):

$$W = 64.852.987 \text{ lx} \cdot \text{min}$$

According to figure 3: $1 \text{ W/m}^2 = 0,13 \text{ klx} = 130 \text{ lx}$ resulting in:

$$W = \frac{64.852.987 \text{ lx} \cdot \text{min}}{130 \text{ lx}} \text{ W/m}^2 = 498.869 \text{ W} \cdot \text{min/m}^2 \text{ per day}$$

Per second and per area of the solar cell:

$$E = 498.869 \text{ W} \cdot 60 \text{ s/m}^2 \cdot 0,001575 \text{ m}^2 = 47.143 \text{ J [W} \cdot \text{s] per day}$$

$$E = 13,1 \frac{\text{Wh}}{\text{d}}$$

Capacitor:

$$E = \frac{1}{2} C U^2 \Rightarrow C = \frac{2 \cdot E}{U^2} = \frac{2 \cdot 47.143 \text{ V} \cdot \text{A} \cdot \text{s}}{(5,5 \text{ V})^2} = \frac{2 \cdot 47.143 \text{ A} \cdot \text{s}}{30,25 \text{ V}} = 3117 \text{ F (with Faktor 2: 6234 F)}$$

A lot!!



15 F @ 5,5 V
(1,6 x 1,6 x 3,3 cm)



500 F @ 16 V
(42 x 16 x 7 cm)



4000 F @ 2,5 V
(15 cm x ø 6,5 cm)

(all: mouser.de)



Energy contained in a standard 18650 Li-Ion rechargeable battery:

$$E = 2000 \text{ mAh} \cdot 3,7 \text{ V (av. voltage)} = \underline{7,2 \text{ Wh}} \rightarrow 2 \times 18650 \text{ would be sufficient (better choice!)}$$

Task 2: Li-Ion Button Cell

A Li-ion button cell has an open circuit voltage of 3.811 V. If a load of $300\ \Omega$ is connected, the voltage drops to 3.795 V.

- a) What is the internal resistance of the button cell?

Solution

$$U_0 = 3,811\text{ V}$$

$$U_L = 3,795\text{ V @ } 300\ \Omega$$

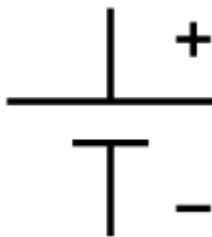
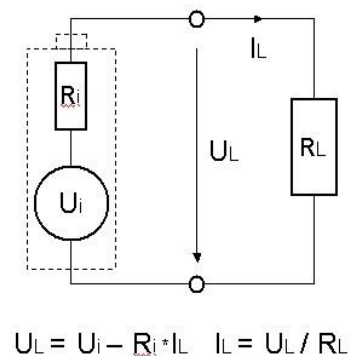
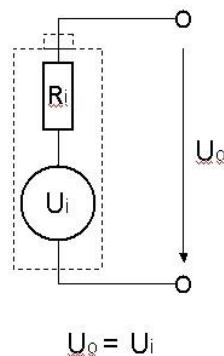
$$\frac{R_i}{R_L} = \frac{(U_0 - U_L)}{U_L} \Rightarrow R_i = \frac{(U_0 - U_L)}{U_L} \cdot R_L = \frac{(3,811\text{ V} - 3,795\text{ V})}{3,795\text{ V}} \cdot 300\ \Omega = \underline{1,265\ \Omega}$$

$$I_L = \frac{U_L}{R_L} = \frac{3,795\text{ V}}{300\ \Omega} = 0,01265\text{ A}$$

- b) Draw an equivalent circuit diagram!

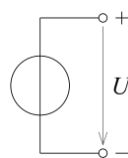
Solution

Circuit symbol:

Equivalent Circuit Diagrams

$$R_i = R_L \cdot \frac{U_0 - U_L}{U_L}$$

U_i as ideal voltage source



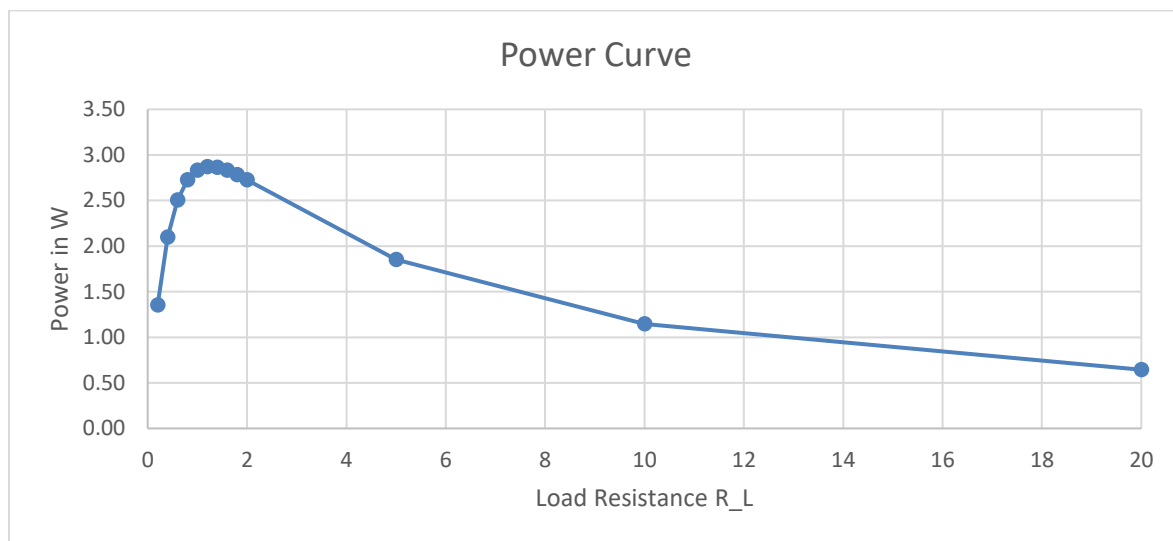
c) Calculate the power at the load resistance for the following values:

0,2	0,4	0,6	0,8	1	1,2	1,4	1,6	1,8	2	5	10	20	50	100	1000	Ω
																W

Solution

R_L	0,2	0,4	0,6	0,8	1,0	1,2	1,4	1,6	1,8	2,0	5,0	10	20	50	100	1000
P_L	1,35	2,10	2,51	2,73	2,83	2,87	2,86	2,83	2,78	2,73	1,85	1,14	0,64	0,28	0,14	0,01

d) Sketch the power curve!



e) What happens at very high load resistances?

Solution

No or hardly any current flows over the load resistance

➔ *No power transmission*

Task 3: SuperCap

Calculate how long a SuperCap with a capacity of 0.3 F / 5.5 V must be charged by a solar cell at a constant global radiation of 300 W/m² so that an MSP430FR263 (16 Mhz → approx. 420 μ A) can be switched on at full "load" and operated for 60 s. Calculate this for the case of an ideal transformer operating at the optimum operating point of the solar cell and neglect the losses of the capacitor.

The technical data on the "Power On" of the MSP in Figure 4 and the I-V and P-V characteristics of the solar cell can be found in Figure 5.

Table 5-2. PMM, SVS and BOR

over recommended ranges of supply voltage and operating free-air temperature (unless otherwise noted)

PARAMETER	TEST CONDITIONS	MIN	TYP	MAX	UNIT
$V_{BOR, safe}$	Safe BOR power-down level ⁽¹⁾	0.1			V
$t_{BOR, safe}$	Safe BOR reset delay ⁽²⁾	10			ms
$I_{SVSH, AM}$	SVS _H current consumption, active mode	$V_{CC} = 3.6$ V		1.5	μ A
$I_{SVSH, LPM}$	SVS _H current consumption, low-power modes	$V_{CC} = 3.6$ V	240		nA
V_{SVSH-}	SVS _H power-down level	1.71	1.80	1.86	V
V_{SVSH+}	SVS _H power-up level	1.74	1.89	1.99	V
$V_{SVSH, hys}$	SVS _H hysteresis		80		mV
$t_{PD, SVSH, AM}$	SVS _H propagation delay, active mode			10	μ s
$t_{PD, SVSH, LPM}$	SVS _H propagation delay, low-power modes			100	μ s
$V_{REF, 1.2V}$	1.2-V REF voltage ⁽³⁾	1.158	1.20	1.242	V

(1) A safe BOR can be correctly generated only if DVCC drops below this voltage before it rises.

(2) When an BOR occurs, a safe BOR can be correctly generated only if DVCC is kept low longer than this period before it reaches V_{SVSH+} .

(3) This is a characterized result with external 1-mA load to ground from -40°C to 85°C.

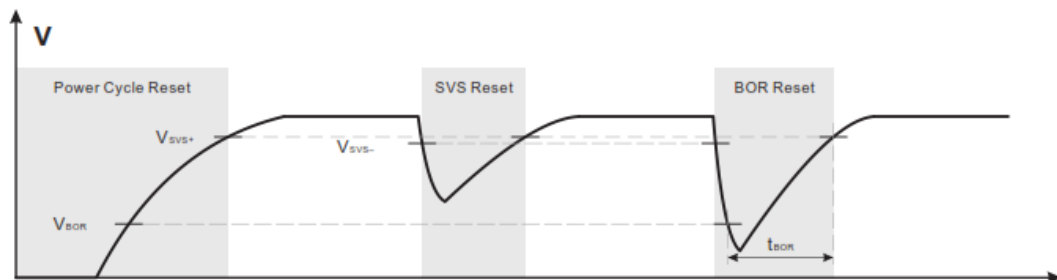
**Figure 5-4. Power Cycle, SVS, and BOR Reset Conditions**

Figure 4 - Technical data MSP430 - Power Cycles

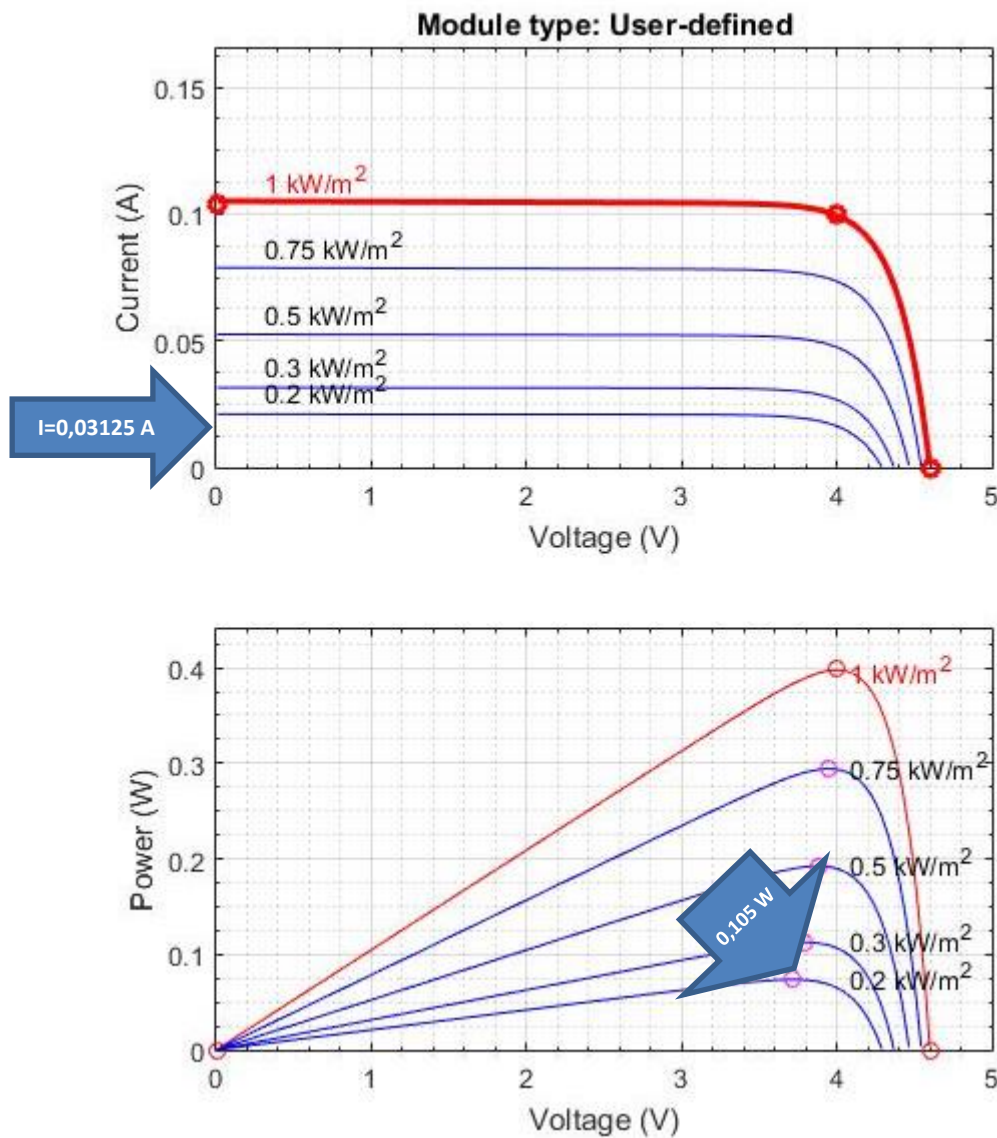


Figure 5 - I-V and P-V curves of a sample solar cell

Solution

1. Calculation of the energy requirement of the MSP for 60 seconds:

$$t = 60 \text{ s}$$

$$P_{MSP} = I_{MSP} \cdot U_{MSP} = 420 \mu\text{A} \cdot 1,89 \text{ V} = 794 \mu\text{W}$$

$$E_{MSP} = P_{MSP} \cdot t = 794 \mu\text{W} \cdot 60 \text{ s} = 47,6 \text{ mJ}$$

2. Calculation of the necessary voltage difference in the capacitor:

$$E = \frac{1}{2} C \cdot (U_2^2 - U_{MSP}^2) \Rightarrow U_2 = \sqrt{\frac{2 \cdot E}{C} + U_{MSP}^2} = \sqrt{\frac{2 \cdot 47,6 \text{ mJ}}{0,3 \text{ F}} + (1,89 \text{ V})^2} = 1,972 \text{ V}$$

3. with MPPT and constant operation of the solar cell:

$$E = P \cdot t \Rightarrow t = \frac{E}{P} = \frac{\frac{1}{2} \cdot 0,3 \text{ F} \cdot (1,972 \text{ V})^2}{0,105 \text{ W}} = 5,56 \text{ s}$$

4. without MPPT: Charge capacitor to 1.972 V.

The solar cell delivers a constant current at $G = 300 \text{ W/m}^2$ of $I = 0,03125 \text{ A}$.

The required charge is $Q = C \cdot U = 0,3 \text{ F} \cdot 1,972 \text{ V} = 0,592 \text{ As}$

This results in the charging time $Q = I \cdot t \Rightarrow t = \frac{Q}{I} = \frac{0,592 \text{ As}}{0,03125 \text{ A}} = 19 \text{ s}$